Yedluri Harshil Rathan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.21
SRI CHAITANYA TECHNO SCHOOL MANGALORE	Class XII (CBSE)	2024	87.40%
KALPATARU CENTRAL SCHOOL	Class X (CBSE)	2022	94.40%

WORK EXPERIENCE

Hardware Intern - EigenAxis.ai

- Developing an **FPGA-based hardware architecture for SLMs**, focusing on compute engines, memory hierarchy, and efficient execution of AI workloads.

Design Intern - Qulabs | Quantum AI

1.**Power Distribution Network |** *Power Electronics*

- Designed a multi-output power supply converting a **24 V** input into regulated **12 V/2 A, 5 V/2 A, 2 V/3 A,** and **30 V/0.1 A** rails using synchronous **buck and boost** converters.
- Designed **SMPS** circuits around dedicated power ICs including **LM5148, LM60430,** and **LM5157**, selecting inductors, capacitors, feedback components, and other external components for each converter stage
- Used **TI WEBENCH Designer** for simulation. Optimised the power stages to achieve a target output **voltage ripple** below **100 mV** across the regulated rails.

2.**Dual Polarity Power Supply |** *Power Electronics*

- Designed a **dual-polarity switching power supply** generating independently regulated **+5 to +12 V / 3 A** and **5 to 12 V / 2 A** output rails with configurable voltage selection.
- Designed switching power stages incorporating **inductors, MOSFETs, diodes, feedback networks, compensation components, and LC output filters** for regulated operation.
- Implemented **DIP switch configurable** feedback and **resistor networks** for multiple output voltage settings, achieving measured **output ripple below 100 mV.**

SKILLS

- **Programming Languages :** C, Python, Verilog, Matlab
- **Development Tools :** Github, Linux, LateX, VSCode, Gerrit, Jira, Excel
- **Simulation Tools :** LTSpice, Altium Designer, Intel Quartus Prime, KiCAD, Arduino IDE, Vivado, STM32CubeIDE

RELEVANT COURSES

- **Core Courses :** Analog Electronics, Digital Systems, Circuits & Network Analysis, Signal Processing, Control systems
- **Math Courses :** Machine Learning, Matrix theory, Differential equations, Probability, Scientific Programming
- **Lab Courses :** Electrical Circuits Lab, Digital Systems Lab, Electronic Devices Lab, PCB Design**,** Signal Processing Lab

PROJECTS

CMOS Digital Design and Layout | VLSI | Github

- Designed and simulated **Inverter**, **2/3-input NAND** and **NOR gates** using **180 nm** CMOS models in **LTspice.**
- Optimized NMOS/PMOS sizing and evaluated **rise, fall times**, **propagation delay**, and **capacitive loading** effects.
- Implemented CMOS schematics and physical layouts in **Electric VLSI Tool** with DRC and RC extraction

5-Stage Pipelined RISC-V Processor | RISC | Github

- Designed and implemented a **32-bit** RISC-V processor, evolving from a **single-cycle** to a **5-stage pipelined** architecture.
- Developed the **Program Counter, Register File, ALU, Control Unit, Immediate Generator**, and pipeline registers.
- Implemented **Instruction Memory, Data Memory**, and **datapath control** across all five pipeline stages.
- Designed a Forwarding Unit with forwarding multiplexers to resolve **ALU data hazards** and **improve pipeline throughput.**

STM32-Based Diode V-I Characterization System | *Analog Circuit Design* | *Github*

- Designed and routed a **4 Layer STM32**-based **PCB** integrating **12-bit DAC, 12-bit ADC, USB-UART**
- Implemented embedded firmware in STM32CubeIDE for DAC sweeping, synchronized **ADC sampling**, and **data acquisition.**
- Designed the analog front-end using MCP6002 **op-amp**, **current-sensing resistor**, and diode interface.
- Integrated CP2102 **USB-UART** for real-time transfer of voltage and current measurements to a PC.
- Generated **forward and reverse** diode V-I characteristics by processing experimental data in Python.

Pipelined FFT & FIR Filter on FPGA | Verilog | Github

- Designed and implemented fixed-point **FIR and FFT digital filters** in Verilog, **comparing pipelined and non-pipelined** architectures for performance and **resource utilization.**
- Synthesized and deployed the DSP designs on the **Intel DE10-Nano** FPGA using **Quartus Prime.**
- Verified functionality using simulation testbenches and SignalTap, performing RTL debugging and timing verification.

8 bit Computer | Bus Architecture

- Designed and built an 8-bit computer using discrete logic ICs, **ALU**, registers, **RAM**, control unit, clock, and **shared data bus.**
- Implemented **instruction execution**, **memory interfacing**, bus arbitration, and control logic for arithmetic and data transfer

PORS

- **Teaching Assistant** - Guided students on projects, assignments, clarified doubts, conducted weekly viva sessions and tutorials on software tools (Python, LaTeX, Github), worked under Prof Dr. GVV Sharma
- **Sunshine Management Team** - Contributed to media coverage and student activities as part of the college well-being club